

Untitled

March 10, 2025

```
[10]: import pandas as pd
import numpy as np
from pyspark.sql import SparkSession

from sklearn.datasets import load_iris
```

```
[19]: from pyspark.sql.functions import *
```

```
[11]: spark = SparkSession.builder.appName("BasicTransformations&Actions").
    ↪getOrCreate()
```

```
[12]: sc = spark.sparkContext
```

```
[13]: iris = spark.read.csv('iris.csv', header=True, inferSchema=True)
```

```
[14]: iris
```

```
[14]: DataFrame[sepal length (cm): double, sepal width (cm): double, petal length
      (cm): double, petal width (cm): double, target: string]
```

```
[15]: iris.describe()
```

```
[15]: DataFrame[summary: string, sepal length (cm): string, sepal width (cm): string,
      petal length (cm): string, petal width (cm): string, target: string]
```

```
[16]: iris.printSchema()
```

```
root
 |-- sepal length (cm): double (nullable = true)
 |-- sepal width (cm): double (nullable = true)
 |-- petal length (cm): double (nullable = true)
 |-- petal width (cm): double (nullable = true)
 |-- target: string (nullable = true)
```

```
[17]: iris.show(4)
```

```
+-----+-----+-----+-----+-----+
--+
```

```
|sepal length (cm)|sepal width (cm)|petal length (cm)|petal width (cm)|
target|
+-----+-----+-----+-----+-----+
--+
|          5.1|          3.5|          1.4|          0.2|Iris-
setosa|
|          4.9|          3.0|          1.4|          0.2|Iris-
setosa|
|          4.7|          3.2|          1.3|          0.2|Iris-
setosa|
|          4.6|          3.1|          1.5|          0.2|Iris-
setosa|
+-----+-----+-----+-----+
--+
only showing top 4 rows
```

```
[21]: only_setosa = iris.filter(col("target") == 'Iris-setosa')
only_setosa.show()
```

```
+-----+-----+-----+-----+-----+
--+
|sepal length (cm)|sepal width (cm)|petal length (cm)|petal width (cm)|
target|
+-----+-----+-----+-----+-----+
--+
|          5.1|          3.5|          1.4|          0.2|Iris-
setosa|
|          4.9|          3.0|          1.4|          0.2|Iris-
setosa|
|          4.7|          3.2|          1.3|          0.2|Iris-
setosa|
|          4.6|          3.1|          1.5|          0.2|Iris-
setosa|
|          5.0|          3.6|          1.4|          0.2|Iris-
setosa|
|          5.4|          3.9|          1.7|          0.4|Iris-
setosa|
|          4.6|          3.4|          1.4|          0.3|Iris-
setosa|
|          5.0|          3.4|          1.5|          0.2|Iris-
setosa|
|          4.4|          2.9|          1.4|          0.2|Iris-
setosa|
|          4.9|          3.1|          1.5|          0.1|Iris-
setosa|
|          5.4|          3.7|          1.5|          0.2|Iris-
setosa|
```

```

|          4.8|          3.4|          1.6|          0.2|Iris-
setosa|
|          4.8|          3.0|          1.4|          0.1|Iris-
setosa|
|          4.3|          3.0|          1.1|          0.1|Iris-
setosa|
|          5.8|          4.0|          1.2|          0.2|Iris-
setosa|
|          5.7|          4.4|          1.5|          0.4|Iris-
setosa|
|          5.4|          3.9|          1.3|          0.4|Iris-
setosa|
|          5.1|          3.5|          1.4|          0.3|Iris-
setosa|
|          5.7|          3.8|          1.7|          0.3|Iris-
setosa|
|          5.1|          3.8|          1.5|          0.3|Iris-
setosa|
+-----+-----+-----+-----+
--+
only showing top 20 rows

```

```
[23]: avg_sepal_lenght = iris.groupBy(col("target")).agg(avg(col("`sepal length_
↪(cm)`")).alias("avg_sepal_lenght"))
```

```
[24]: avg_sepal_lenght.show()
```

```

[Stage 7:>                                     (0 + 1) /
1]

+-----+-----+
|      target| avg_sepal_lenght|
+-----+-----+
|Iris-versicolor|          5.936|
| Iris-virginica|6.587999999999998|
|   Iris-setosa|5.005999999999999|
+-----+-----+

```

```
[26]: # Create a new column 'sepal_area' (sepal_length * sepal_width)
iris.withColumn("sepal_area", col("`sepal length (cm)`" * col("`sepal width_
↪(cm)`")).show()
```

```

+-----+-----+-----+-----+
--+-----+
|sepal length (cm)|sepal width (cm)|petal length (cm)|petal width (cm)|
target|      sepal_area|

```

```

+-----+-----+-----+-----+
--+-----+
|          5.1|          3.5|          1.4|          0.2|Iris-
setosa|17.849999999999998|
|          4.9|          3.0|          1.4|          0.2|Iris-
setosa|14.700000000000001|
|          4.7|          3.2|          1.3|          0.2|Iris-
setosa|15.040000000000001|
|          4.6|          3.1|          1.5|          0.2|Iris-
setosa|          14.26|
|          5.0|          3.6|          1.4|          0.2|Iris-
setosa|          18.0|
|          5.4|          3.9|          1.7|          0.4|Iris-
setosa|21.060000000000002|
|          4.6|          3.4|          1.4|          0.3|Iris-
setosa|15.639999999999999|
|          5.0|          3.4|          1.5|          0.2|Iris-
setosa|          17.0|
|          4.4|          2.9|          1.4|          0.2|Iris-
setosa|          12.76|
|          4.9|          3.1|          1.5|          0.1|Iris-
setosa|15.190000000000001|
|          5.4|          3.7|          1.5|          0.2|Iris-
setosa|19.980000000000004|
|          4.8|          3.4|          1.6|          0.2|Iris-
setosa|          16.32|
|          4.8|          3.0|          1.4|          0.1|Iris-
setosa|14.399999999999999|
|          4.3|          3.0|          1.1|          0.1|Iris-
setosa|12.899999999999999|
|          5.8|          4.0|          1.2|          0.2|Iris-
setosa|          23.2|
|          5.7|          4.4|          1.5|          0.4|Iris-
setosa|25.080000000000002|
|          5.4|          3.9|          1.3|          0.4|Iris-
setosa|21.060000000000002|
|          5.1|          3.5|          1.4|          0.3|Iris-
setosa|17.849999999999998|
|          5.7|          3.8|          1.7|          0.3|Iris-
setosa|          21.66|
|          5.1|          3.8|          1.5|          0.3|Iris-
setosa|          19.38|
+-----+-----+-----+-----+
--+-----+
only showing top 20 rows

```

```
[27]: # 5. Sorting: Order rows by petal_length in descending order
sortedDF = iris.orderBy(col("`petal length (cm)`").desc())
sortedDF.show(5)
```

```
+-----+-----+-----+-----+-----+
-----+
|sepal length (cm)|sepal width (cm)|petal length (cm)|petal width (cm)|
target|
+-----+-----+-----+-----+-----+
-----+
|          7.7|          2.6|          6.9|          2.3|Iris-
virginica|
|          7.7|          3.8|          6.7|          2.2|Iris-
virginica|
|          7.7|          2.8|          6.7|          2.0|Iris-
virginica|
|          7.6|          3.0|          6.6|          2.1|Iris-
virginica|
|          7.9|          3.8|          6.4|          2.0|Iris-
virginica|
+-----+-----+-----+-----+-----+
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only showing top 5 rows
```